

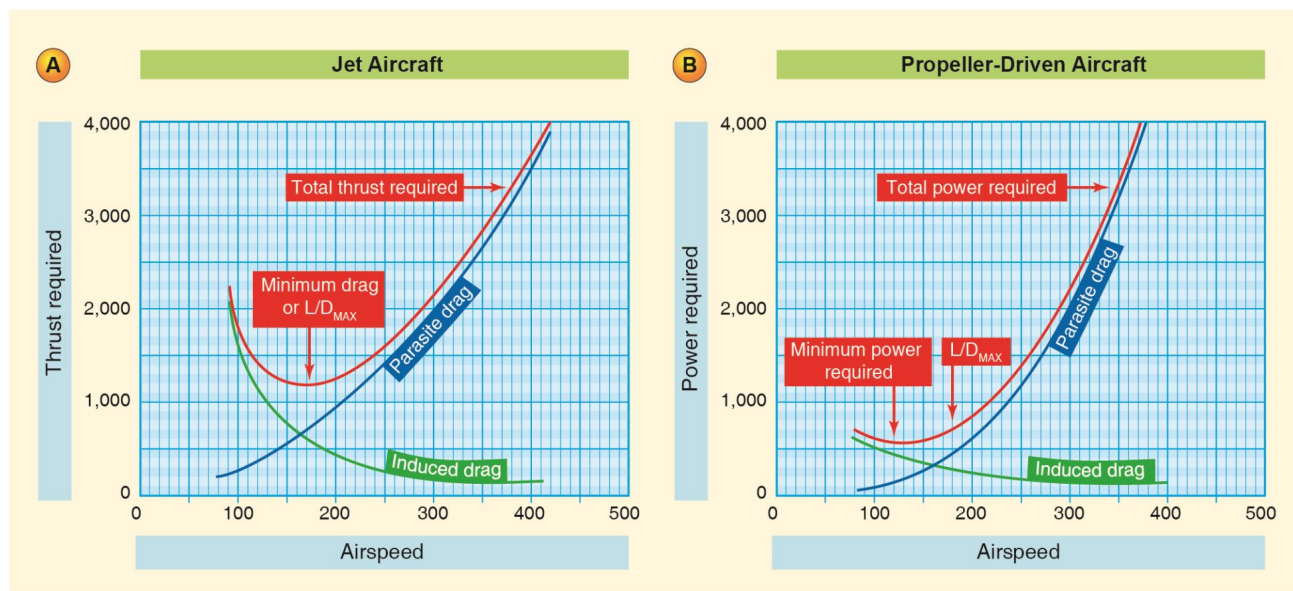


Figure 16-7. Thrust and power required curves (jet aircraft vs. propeller-driven aircraft).

Stalls

The stalling characteristics of the swept wing jet airplane can vary considerably from those of the normal straight wing airplane. The greatest difference noticeable to the pilot is the lift developed vs. angle of attack. An increase in angle of attack of the straight wing produces a substantial and constantly increasing lift vector up to its maximum coefficient of lift, and soon thereafter flow separation (stall) occurs with a rapid deterioration of lift.

By contrast, the swept wing produces a much more gradual buildup of lift with a less well-defined maximum coefficient. This less-defined peak also means that a swept wing may not have as dramatic a loss of lift at angles of attack beyond its maximum lift coefficient. However, these high-lift conditions are accompanied by high drag, which may result in a high rate of descent. [Figure 16-8]

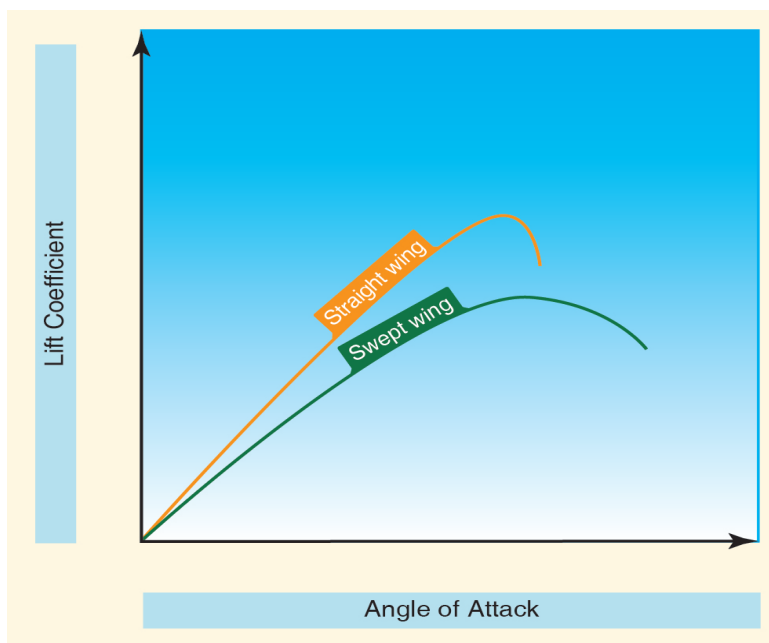


Figure 16-8. Stall versus angle of attack—swept wing versus straight wing.

If a simple, straight-wing airplane's airfoil is swept, a natural tendency arises that it will stall at the wing tips first. This is because the boundary layer tends to flow spanwise toward the tips. [Figure 16-9] The tendency for tip stall allowing the center of lift to move forward is greatest when wing sweep and taper are combined. To discourage a swept wing from stalling at the wingtips, manufacturers modify the wing spanwise with twist, changes in airfoil section, inclusion of vortex generators, or a combination of those modifications. This helps a pilot retain roll control initially if a stall is entered inadvertently.

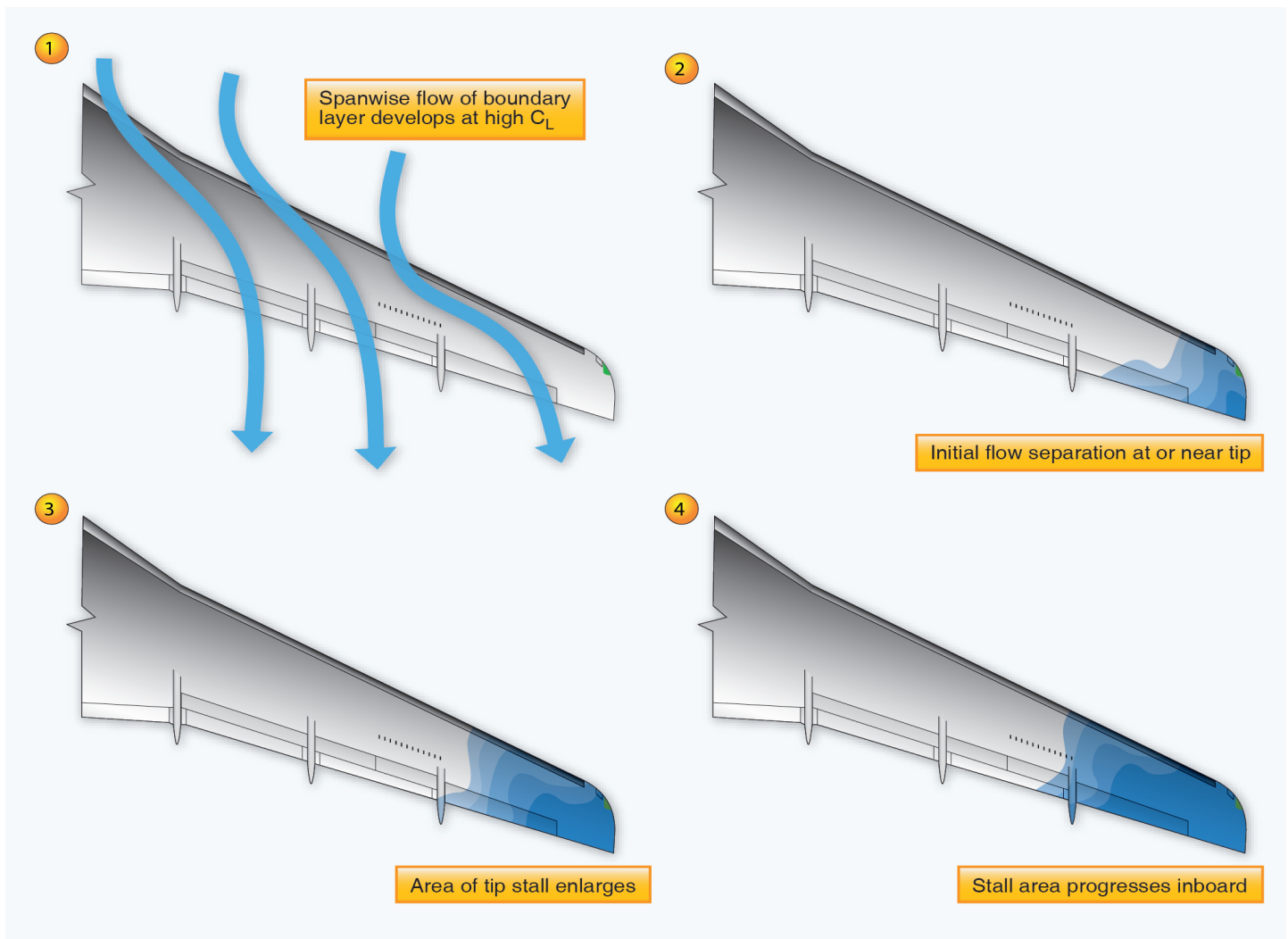


Figure 16-9. *Unmodified swept wing stall characteristics.*

Some T-tail configurations are prone to so-called deep stalls where the tail can become immersed in the wing wake at very high angles of attack and lose effectiveness. Such a situation can be accompanied by a high-rate of descent. Since high angles of attack can occur at any pitch attitude, even a pitch attitude with the nose below the horizon, it may seem counterintuitive in such a situation to take the appropriate recovery action, which is to push the nose down even further.

Deep stalls may be unrecoverable. Fortunately, they are easily avoided as long as published limitations are observed. On those airplanes susceptible to deep stalls (not all swept or tapered wing airplanes are), sophisticated stall warning systems such as stick shakers are standard equipment. A stick pusher (if installed), as its name implies, acts to automatically reduce the airplane's AOA before the airplane reaches a dangerous stall condition, or it may aid in recovering the airplane from a stall if an airplane's natural aerodynamic characteristics do so weakly. Pilots should avoid situations that would activate a stick pusher when close to the ground.

Pilots undergoing training in jet airplanes are taught to recover at the first indication of an impending stall instead of going beyond those initial cues and into a full stall. Normally, this is indicated by aural stall warning devices, annunciators, or activation of the airplane's stick shaker. Stick shakers normally activate around 107 percent of the actual stall speed. In response to a stall warning, the proper action is for the pilot to apply a nose-down input until the stall warning stops (pitch trim may be necessary). Then, the wings are rolled level, followed by adjusting thrust to return to normal flight. The elapsed time will be small between these actions, particularly at low altitude where significant available thrust exists. It is important to understand that reducing AOA eliminates the stall, but added thrust will allow the descent to be stopped once the wing is flying again. Note that airplanes without vortex generators may stall with little to no buffet.

At high altitudes the stall recovery technique is the same. A pilot will need to reduce the AOA by lowering the nose until the stall warning stops. However, after the AOA has been reduced to where the wing is again developing efficient lift, the airplane will still likely need to accelerate to a desired airspeed. At high altitudes where the available thrust is significantly less than at lower altitudes, recovery may require significant pitch down to regain airspeed. As such, several thousand feet or more of altitude loss may occur during the recovery. The above discussion covers most airplanes; however, the stall recovery procedures for a particular make and model airplane may differ, as recommended by the manufacturer, and are contained in the FAA-approved Airplane Flight Manual for that airplane.

Drag Devices

Jet airplanes have higher glide ratios than piston-powered airplanes. Due to their low drag design, jets take more time and distance to descend or reduce speed. Therefore, jet airplanes are often equipped with drag devices, such as spoilers and speed brakes.

The primary purpose of spoilers is to spoil lift. The most common type of spoiler consists of one or more rectangular plates that lie flush with the upper surface of each wing. They are installed approximately parallel to the lateral axis of the airplane and are hinged along the leading edges. When deployed, spoilers deflect up against the relative wind, which interferes with the flow of air about the wing. [Figure 16-10] This both spoils lift and increases drag. Spoilers are usually installed forward of the flaps but not in front of the ailerons so as not to interfere with roll control. Some aircraft use spoilers to augment roll control.



Figure 16-10. Spoilers.

When flight and ground spoilers are deployed after landing, most of the wing's lift is destroyed. This action transfers the airplane's weight to the landing gear so that the wheel brakes are more effective. A secondary beneficial effect of deploying spoilers on landing is that they create considerable drag, adding to the overall aerodynamic braking.

The primary purpose of speed brakes is to produce drag. Spoilers may also serve as speed brakes, or they may be panels attached to the fuselage. Deploying speed brakes results in a rapid decrease in airspeed and/or an increased rate of descent. Typically, speed brakes can be deployed at any time during flight. There is usually a certain amount of noise and buffeting associated with the use of speed brakes, along with an obvious penalty in fuel consumption. Pilots can minimize the use of speed brakes with proper descent and approach planning. Procedures for the use of spoilers and/or speed brakes in various situations are contained in the FAA-approved AFM for the particular airplane.

Thrust Reversers

Jet airplanes have high kinetic energy during the landing roll because of weight and speed. This energy is difficult to dissipate because a jet airplane has low drag with the nose-wheel on the ground, and the engines continue to produce forward thrust with the power levers at idle. While wheel brakes serve as the primary means to stop the airplane, reverse thrust, when available, assists in deceleration.

Certain thrust reverser designs effectively reverse the flow of the exhaust gases. The flow does not completely reverse. Typically, the final path of the exhaust gases is about 45° from straight ahead. This, together with the losses from the flow paths, reduces reverse thrust efficiency. If the pilot uses less than maximum rpm in reverse, the reverse thrust is further reduced.

Normally, a jet engine has one of two types of thrust reversers: a target reverser or a cascade reverser. [Figure 16-11] Target reversers are simple clamshell doors that swivel from the stowed position at the engine tailpipe to redirect thrust to a more forward direction.

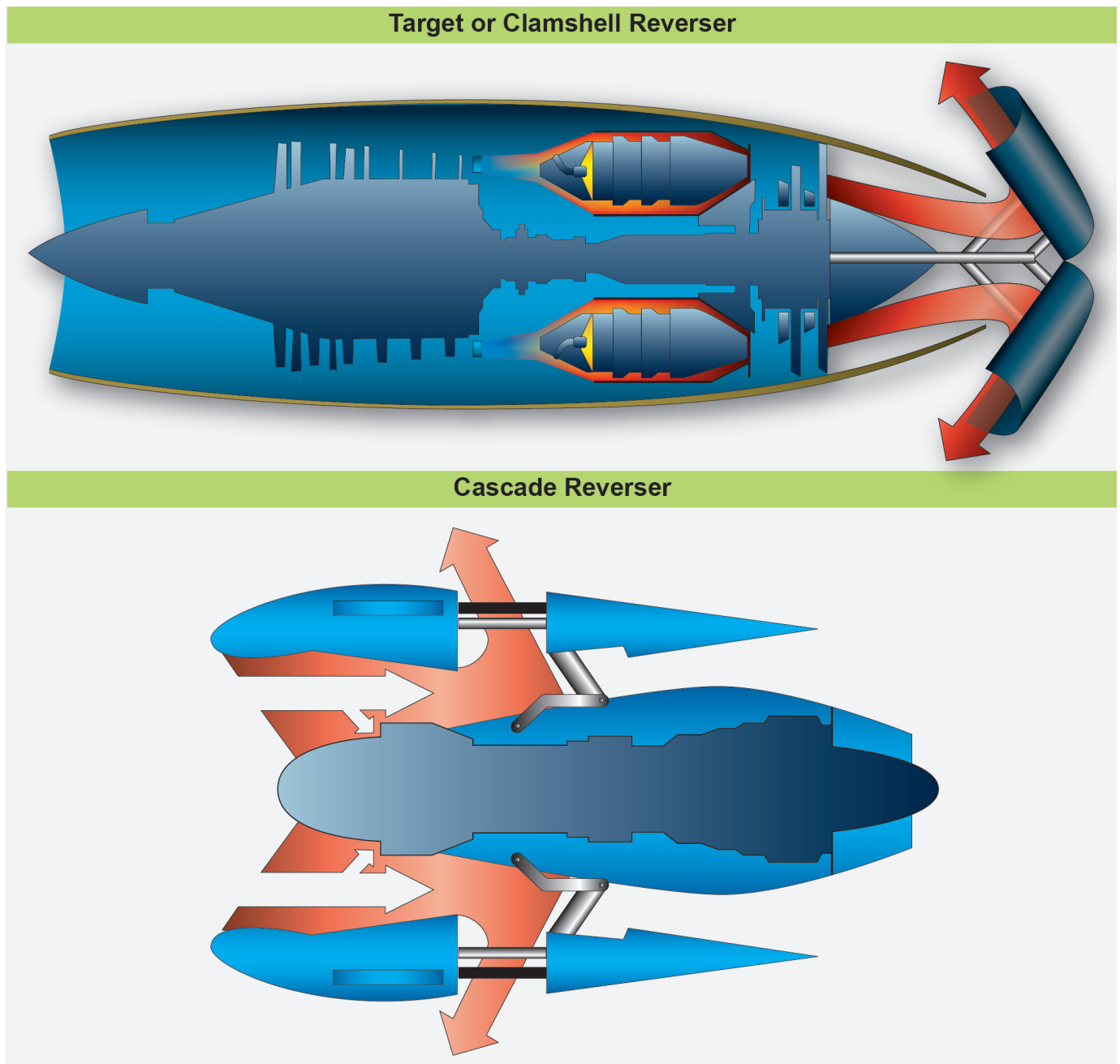


Figure 16-11. Thrust reversers.

Cascade reversers are normally found on turbofan engines and are often designed to reverse only the fan air portion. Blocking doors in the shroud obstruct forward fan thrust and redirect it through cascade vanes to generate reverse thrust.

On most installations, the pilot selects reverse thrust with the thrust levers at idle by pulling up the reverse levers to a detent. Doing so positions the reversing mechanisms for operation but leaves the engines at idle rpm. Further upward and backward movement of the reverse levers increases engine power. Reverse is canceled by closing the reverse levers to the idle reverse position, then dropping them fully back to the forward idle position. This last movement selects the stowed position, and the reversers return to the forward thrust position.

Reverse thrust is more effective at high speed than at low speed. For maximum reverse thrust efficiency, the pilot should use it as soon as is prudent after touchdown. The pilot should remember that some airplanes tend to pitch nose-up when reverse is selected on landing and this effect, particularly when combined with the nose-up pitch effect from the spoilers, can cause the airplane to leave the ground again momentarily. On these types, the airplane should be firmly on the ground with the nose-wheel down before reverse is selected. Other types of airplanes have no change in pitch, and reverse idle may be selected after the main gear is down and before the nose-wheel is down. Since reverse thrust may affect directional control, runway surface conditions (e.g., contamination), factor into the use of reverse thrust. Specific procedures for reverse thrust operation for a particular airplane/engine combination are contained in the FAA-approved AFM for that airplane.

There is a significant difference between reverse pitch on a propeller and reverse thrust from a jet engine. Idle reverse on a propeller produces a large amount of drag. On a jet engine, however, selecting idle reverse produces very little reverse thrust. In a jet airplane, the pilot should select reverse, apply reverse thrust as appropriate, and remain within any AFM limitations.

It is essential that pilots understand not only the normal procedures and limitations of thrust reverser use, but also the procedures for coping with uncommanded reverse. While thrust reverser systems are designed to prevent unintentional deployment, an uncommanded or inadvertent deployment of thrust reversers, while airborne, is an emergency. The systems normally contain several lock systems: one to keep reversers from operating in the air, another to prevent operation with the thrust levers out of the idle detent, and/or an “auto-stow” circuit to command reverser stowage any time thrust reverser deployment would be inappropriate, such as during takeoff and while airborne.

Pilot Sensations in Jet Flying

Pilots transitioning into jets may notice these general sensations:

1. response differences
2. increased control sensitivity
3. increased tempo of flight

In some flight conditions, airspeed changes may occur more slowly than in a propeller airplane. At high altitudes, the reduction in available thrust reduces the ability to accelerate. The long spool-up time required from low throttle settings also may affect acceleration. Finally, the clean aerodynamic design of a jet can result in more gradual deceleration when thrust is reduced.

The lack of propeller effects results in less drag at low power settings. Other changes the transitioning pilot should notice include the lack of effective slipstream over the lifting and control surfaces, and the lack of propeller torque effect.

Even though moving the power levers has less effect at low power settings, the pilot should change power settings smoothly. To slow the airplane, the transitioning pilot may also need to learn when to use available drag devices appropriately.

Transitioning pilots should learn power setting management for different situations. Power settings for desired performance vary because of significant changes in airplane weight as fuel is consumed. Therefore, the pilot needs to use a variety of cues to achieve desired performance. For example, airspeed trend information provides feedback for power required.

Power changes may result in a pitching tendency. These characteristics should be noticed and compensated for.

The jet airplane will differ regarding pitch tendencies with the lowering of flaps, landing gear, and drag devices. With experience, the jet airplane pilot will learn to anticipate the pitch change required for a particular operation. Most jet airplanes are equipped with a thumb operated pitch trim button on the control wheel. The usual method of operating the trim button is to apply several small, intermittent applications of trim in the direction desired rather than holding the trim button for longer periods of time, which can lead to overcontrolling.

The variation of pitch attitudes flown in a jet airplane also results from high thrust, flight characteristics of the low aspect ratio, and the swept wing. Flight at higher pitch attitudes requires greater reliance on the flight instruments for airplane control since outside references may be absent. Proficiency in attitude instrument flying, therefore, is essential to successful transition to jet airplane flying.